

The screenshot shows the Eclipse IDE interface with the following components:

- Package Explorer:** Displays the project structure for `student_klh_api`. The `src/main/java` directory is expanded, showing the `com.example.demo` package. The `Product.java` file is selected.
- Outline:** Shows the class hierarchy for `Product` in the `com.example.demo` package. The class `Product` is expanded, showing its attributes (`id`, `name`, `description`, `price`, `quantity`) and methods (`Product()`, `Product(String, String, double, int)`, `getId()`, `setId()`, `getName()`, `setName()`, `getDescription()`, `setDescription()`, `getPrice()`, `setPrice()`, `getQuantity()`, `setQuantity()`).
- Code Editor:** Displays the source code of `Product.java`. The code is as follows:

```
1 package com.example.demo;
2
3 import jakarta.persistence.Entity;
4
5 @Entity
6 @Table(name = "products")
7 public class Product {
8
9     @Id
10    @GeneratedValue(strategy = GenerationType.IDENTITY)
11    private Long id;
12
13    private String name;
14    private String description;
15    private double price;
16    private int quantity;
17
18    public Product() {
19    }
20
21    public Product(String name, String description, double price, int quantity) {
22        this.name = name;
23        this.description = description;
24        this.price = price;
25        this.quantity = quantity;
26    }
27
28    public Long getId() {
29        return id;
30    }
31
32 }
```
- Console:** Shows the message "No consoles to display at this time."
- Boot Dashboard:** Shows the "local" environment.
- Status Bar:** Displays "Writable", "Smart Insert", and the cursor position "12 : 1 : 282".

The screenshot shows the Eclipse IDE interface with the following components:

- Package Explorer (Left):** Displays the project structure for `student_klh_api`. The `src/main/java` directory is expanded, showing the `com.example.demo` package. Inside this package, the following files are listed: `Product.java`, `ProductController.java`, `ProductRepository.java`, `ProductService.java`, `ProductServiceImpl.java`, and `StudentKlhApiApplication.java`.
- Code Editor (Center):** Displays the source code of `Product.java`. The code is as follows:

```
41     return name;
42 }
43
44 public void setName(String name) {
45     this.name = name;
46 }
47
48 public String getDescription() {
49     return description;
50 }
51
52 public void setDescription(String description) {
53     this.description = description;
54 }
55
56 public double getPrice() {
57     return price;
58 }
59
60 public void setPrice(double price) {
61     this.price = price;
62 }
63
64 public int getQuantity() {
65     return quantity;
66 }
67
68 public void setQuantity(int quantity) {
69     this.quantity = quantity;
70 }
71 }
```
- Outline (Right):** Shows the class structure of `Product`. It lists the following attributes: `id : Long`, `name : String`, `description : String`, `price : double`, and `quantity : int`. It also lists the methods: `Product()`, `Product(String, String, double, int)`, `getId() : Long`, `setId(Long) : void`, `getName() : String`, `setName(String) : void`, `getDescription() : String`, `setDescription(String) : void`, `getPrice() : double`, `setPrice(double) : void`, `getQuantity() : int`, and `setQuantity(int) : void`.
- Boot Dashboard (Bottom Left):** Shows a search bar with the text "Type tags, projects, or working set names to match (incl. * and ? wildc...". Below the search bar, there is a section for "local" with a green power icon and the text "1 elements hidden by filter".
- Problems, Javadoc, Declaration, Console (Bottom Right):** These tabs are visible, but the console area is empty, displaying the message "No consoles to display at this time."

ProductController.java — Page 1 of 2

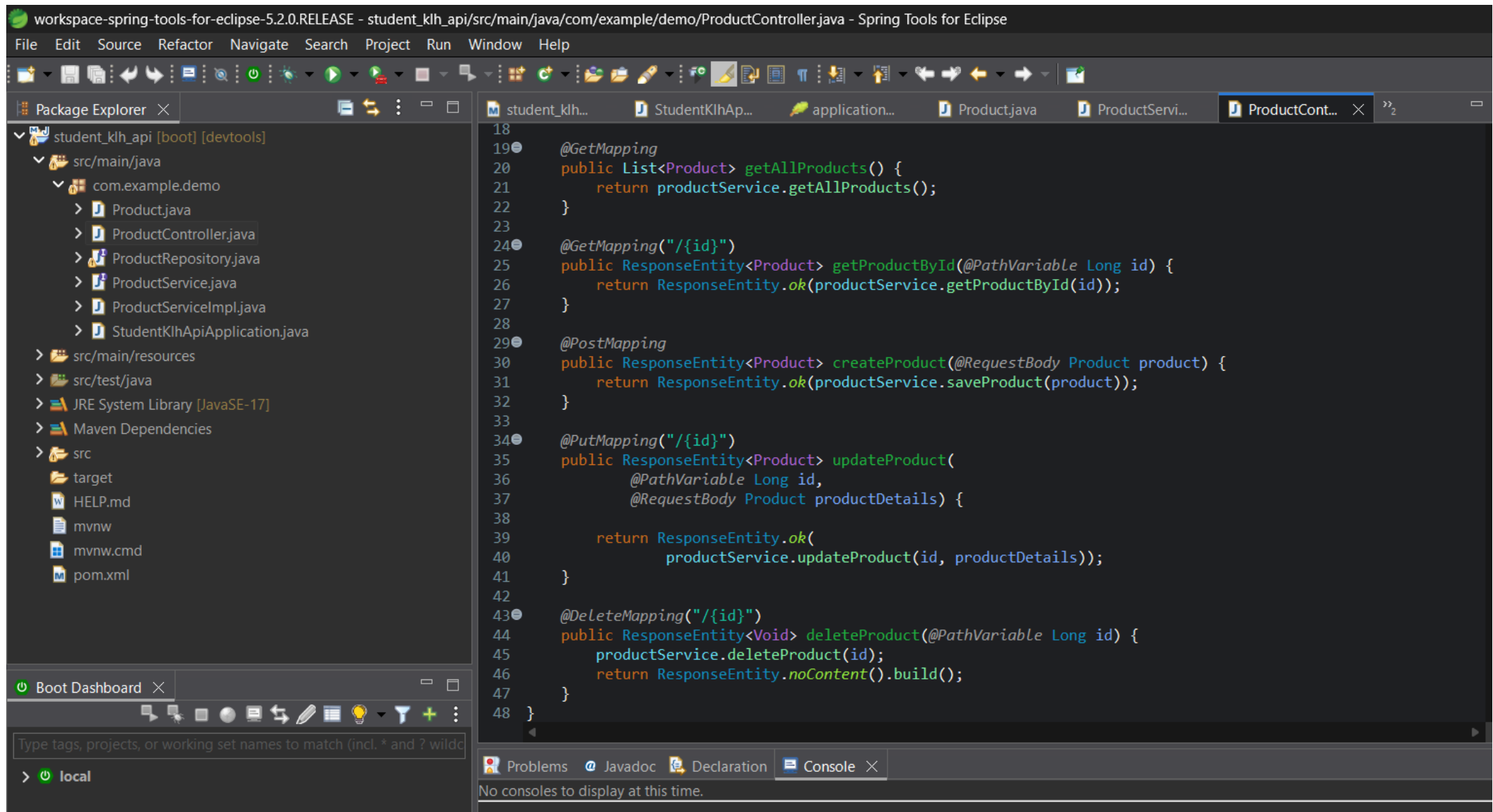
The screenshot shows the Eclipse IDE interface. The top menu bar includes File, Edit, Source, Refactor, Navigate, Search, Project, Run, Window, and Help. The toolbar contains various icons for file operations and development tools. The Package Explorer on the left shows the project structure: student_klh_api [boot] [devtools] with sub-packages src/main/java, src/main/resources, src/test/java, JRE System Library [JavaSE-17], Maven Dependencies, and src. The src/main/java package contains com.example.demo, which includes Product.java, ProductController.java (selected), ProductRepository.java, ProductService.java, ProductServiceImpl.java, and StudentKlhApiApplication.java. The Boot Dashboard at the bottom left shows a search bar and a 'local' button. The main editor displays the code for ProductController.java, which is a REST controller for products. The code includes package, import, annotations, and several methods: constructor, getAllProducts, getProductById, createProduct, and a partially visible putProduct method.

```
1 package com.example.demo;
2
3 import java.util.List;
4
5
6
7
8 @RestController
9 @RequestMapping("/api/products")
10 @CrossOrigin(origins = "http://localhost:5173")
11 public class ProductController {
12
13     private final ProductService productService;
14
15     public ProductController(ProductService productService) {
16         this.productService = productService;
17     }
18
19     @GetMapping
20     public List<Product> getAllProducts() {
21         return productService.getAllProducts();
22     }
23
24     @GetMapping("/{id}")
25     public ResponseEntity<Product> getProductById(@PathVariable Long id) {
26         return ResponseEntity.ok(productService.getProductById(id));
27     }
28
29     @PostMapping
30     public ResponseEntity<Product> createProduct(@RequestBody Product product) {
31         return ResponseEntity.ok(productService.saveProduct(product));
32     }
33
34     @PutMapping("/{id}")
```

Problems Javadoc Declaration Console

No consoles to display at this time.

ProductController.java — Page 2 of 2



workspace-spring-tools-for-eclipse-5.2.0.RELEASE - student_klh_api/src/main/java/com/example/demo/ProductController.java - Spring Tools for Eclipse

File Edit Source Refactor Navigate Search Project Run Window Help

Package Explorer

- student_klh_api [boot] [devtools]
 - src/main/java
 - com.example.demo
 - Product.java
 - ProductController.java
 - ProductRepository.java
 - ProductService.java
 - ProductServiceImpl.java
 - StudentKlhApiApplication.java
 - src/main/resources
 - src/test/java
 - JRE System Library [JavaSE-17]
 - Maven Dependencies
 - src
 - target
 - HELP.md
 - mvnw
 - mvnw.cmd
 - pom.xml

Boot Dashboard

Type tags, projects, or working set names to match (incl. * and ? wildc

local

```
18
19 @GetMapping
20 public List<Product> getAllProducts() {
21     return productService.getAllProducts();
22 }
23
24 @GetMapping("/{id}")
25 public ResponseEntity<Product> getProductById(@PathVariable Long id) {
26     return ResponseEntity.ok(productService.getProductById(id));
27 }
28
29 @PostMapping
30 public ResponseEntity<Product> createProduct(@RequestBody Product product) {
31     return ResponseEntity.ok(productService.saveProduct(product));
32 }
33
34 @PutMapping("/{id}")
35 public ResponseEntity<Product> updateProduct(
36     @PathVariable Long id,
37     @RequestBody Product productDetails) {
38
39     return ResponseEntity.ok(
40         productService.updateProduct(id, productDetails));
41 }
42
43 @DeleteMapping("/{id}")
44 public ResponseEntity<Void> deleteProduct(@PathVariable Long id) {
45     productService.deleteProduct(id);
46     return ResponseEntity.noContent().build();
47 }
48 }
```

Problems Javadoc Declaration Console

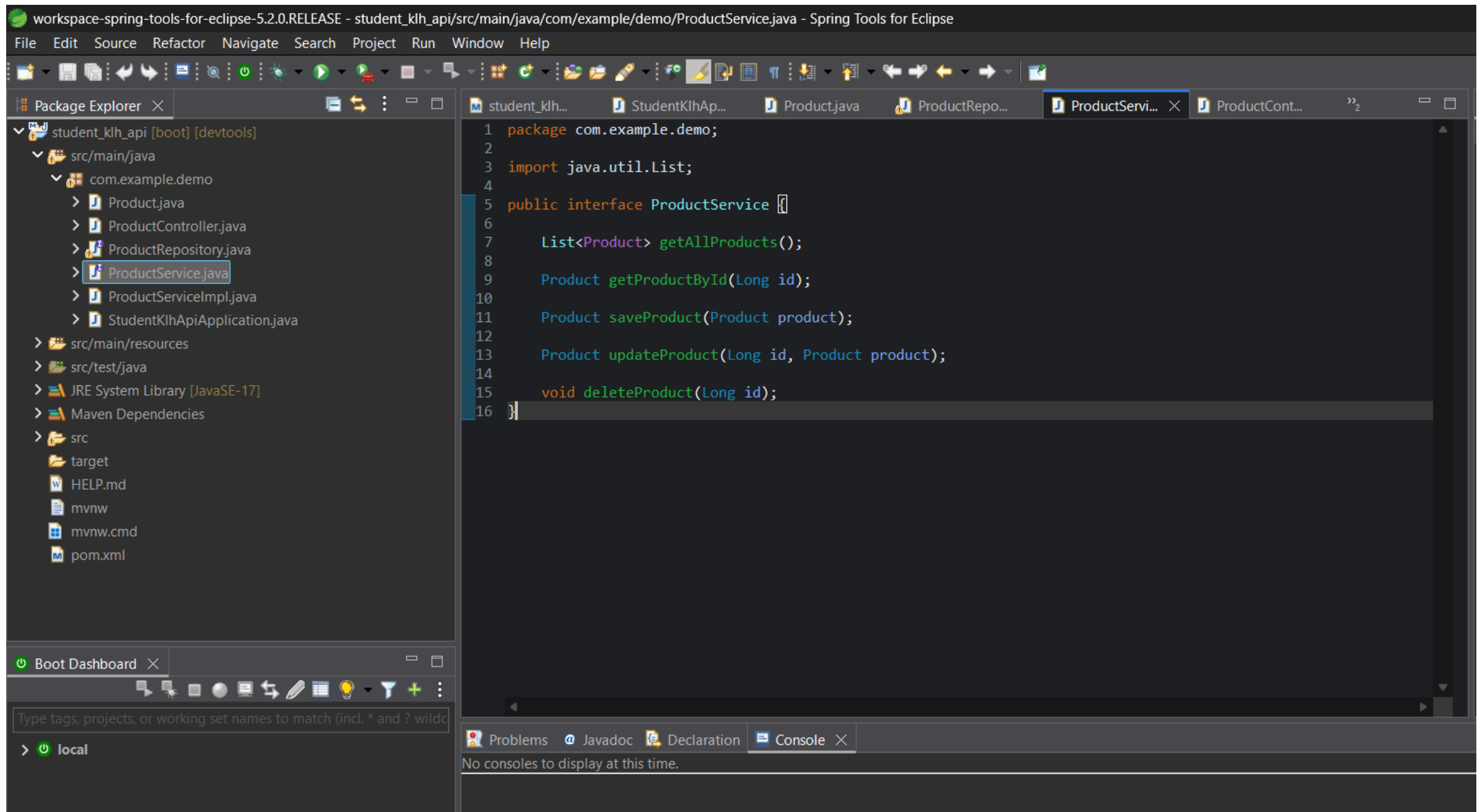
No consoles to display at this time.

ProductRepository.java

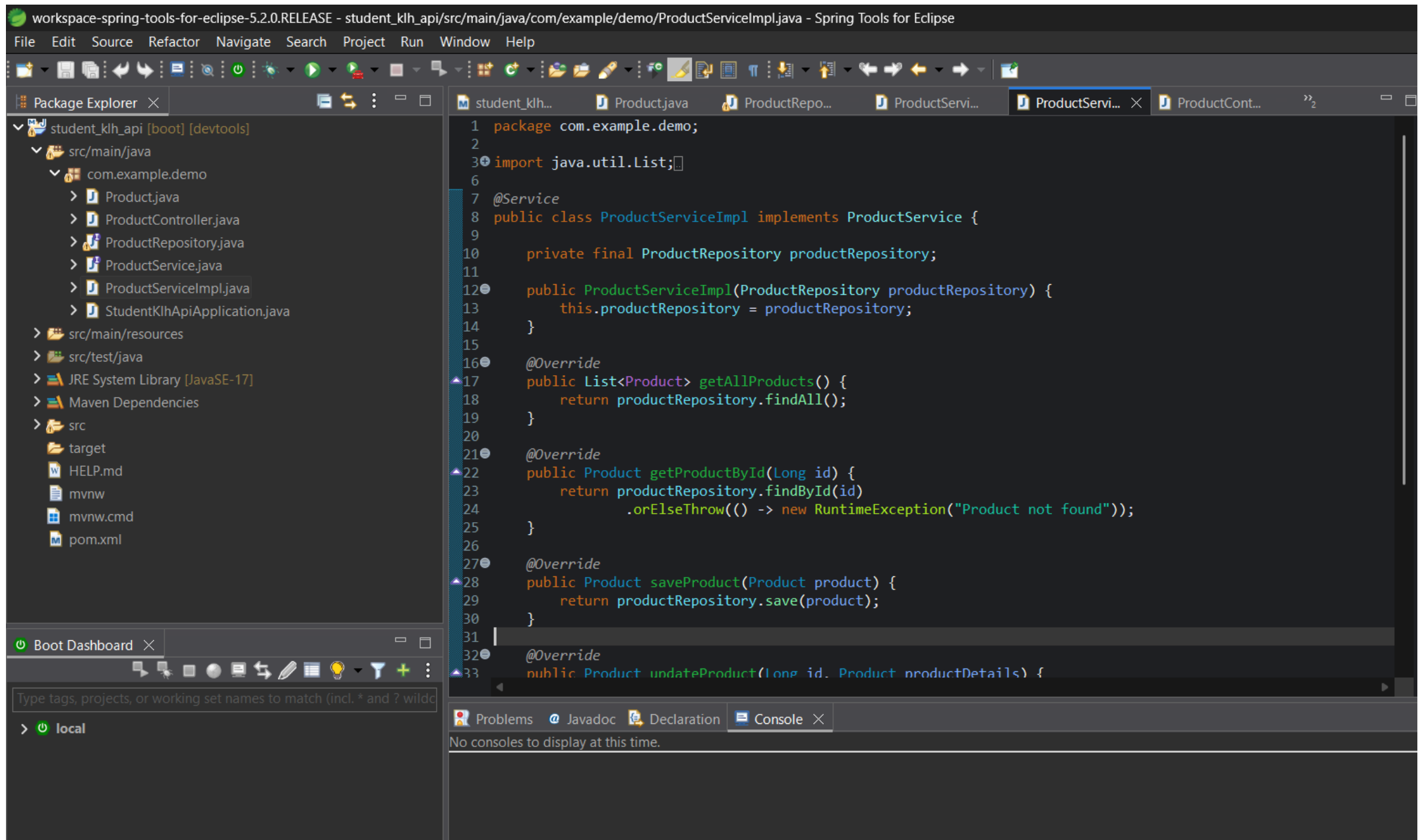
The screenshot shows the Eclipse IDE interface with the following components:

- Title Bar:** workspace-spring-tools-for-eclipse-5.2.0.RELEASE - student_klh_api/src/main/java/com/example/demo/ProductRepository.java - Spring Tools for Eclipse
- Menu Bar:** File, Edit, Source, Refactor, Navigate, Search, Project, Run, Window, Help
- Toolbar:** Standard Eclipse development icons for file operations, editing, and running.
- Package Explorer (Left):**
 - student_klh_api [boot] [devtools]
 - src/main/java
 - com.example.demo
 - Product.java
 - ProductController.java
 - ProductRepository.java** (selected)
 - ProductService.java
 - ProductServiceImpl.java
 - StudentKlhApiApplication.java
 - src/main/resources
 - src/test/java
 - JRE System Library [JavaSE-17]
 - Maven Dependencies
 - src
 - target
 - HELP.md
 - mvnw
 - mvnw.cmd
 - pom.xml
- Boot Dashboard (Bottom Left):**
 - Boot Dashboard
 - Type tags, projects, or working set names to match (incl. * and ? wildcards)
 - local
- Editor (Center):**
 - student_klh... StudentKlhAp... Product.java **ProductRepo...** ProductServi... ProductCont...
 - 1 package com.example.demo;
 - 2
 - 3 import org.springframework.data.jpa.repository.JpaRepository;
 - 4
 - 5
 - 6 @Repository
 - 7 public interface ProductRepository extends JpaRepository<Product, Long> {
 - 8 }
- Problems/Console (Bottom Right):**
 - Problems Javadoc Declaration Console
 - No consoles to display at this time.

ProductService.java



ProductServiceImpl.java — Page 1 of 2



workspace-spring-tools-for-eclipse-5.2.0.RELEASE - student_klh_api/src/main/java/com/example/demo/ProductServiceImpl.java - Spring Tools for Eclipse

File Edit Source Refactor Navigate Search Project Run Window Help

Package Explorer

- student_klh_api [boot] [devtools]
 - src/main/java
 - com.example.demo
 - Product.java
 - ProductController.java
 - ProductRepository.java
 - ProductService.java
 - ProductServiceImpl.java
 - StudentKlhApiApplication.java
 - src/main/resources
 - src/test/java
 - JRE System Library [JavaSE-17]
 - Maven Dependencies
 - src
 - target
 - HELP.md
 - mvnw
 - mvnw.cmd
 - pom.xml

Boot Dashboard

Type tags, projects, or working set names to match (incl. * and ? wildc

> local

```
1 package com.example.demo;
2
3 import java.util.List;
4
5
6
7 @Service
8 public class ProductServiceImpl implements ProductService {
9
10     private final ProductRepository productRepository;
11
12     public ProductServiceImpl(ProductRepository productRepository) {
13         this.productRepository = productRepository;
14     }
15
16     @Override
17     public List<Product> getAllProducts() {
18         return productRepository.findAll();
19     }
20
21     @Override
22     public Product getProductById(Long id) {
23         return productRepository.findById(id)
24             .orElseThrow(() -> new RuntimeException("Product not found"));
25     }
26
27     @Override
28     public Product saveProduct(Product product) {
29         return productRepository.save(product);
30     }
31
32     @Override
33     public Product updateProduct(Long id, Product productDetails) {
```

Problems Javadoc Declaration Console

No consoles to display at this time.

ProductServiceImpl.java — Page 2 of 2

The screenshot shows the Eclipse IDE interface. The top bar indicates the workspace is 'workspace-spring-tools-for-eclipse-5.2.0.RELEASE' and the current file is 'student_klh_api/src/main/java/com/example/demo/ProductServiceImpl.java'. The menu bar includes File, Edit, Source, Refactor, Navigate, Search, Project, Run, Window, and Help. The toolbar contains various icons for file operations and development tools.

The Package Explorer on the left shows the project structure:

- student_klh_api [boot] [devtools]
 - src/main/java
 - com.example.demo
 - Product.java
 - ProductController.java
 - ProductRepository.java
 - ProductService.java
 - ProductServiceImpl.java
 - StudentKlhApiApplication.java
 - src/main/resources
 - src/test/java
 - JRE System Library [JavaSE-17]
 - Maven Dependencies
 - src
 - target
 - HELP.md
 - mvnw
 - mvnw.cmd
 - pom.xml

The main editor displays the code for ProductServiceImpl.java:

```
18     return productRepository.findAll();
19 }
20
21 @Override
22 public Product getProductById(Long id) {
23     return productRepository.findById(id)
24         .orElseThrow(() -> new RuntimeException("Product not found"));
25 }
26
27 @Override
28 public Product saveProduct(Product product) {
29     return productRepository.save(product);
30 }
31
32 @Override
33 public Product updateProduct(Long id, Product productDetails) {
34     Product existingProduct = getProductById(id);
35
36     existingProduct.setName(productDetails.getName());
37     existingProduct.setDescription(productDetails.getDescription());
38     existingProduct.setPrice(productDetails.getPrice());
39     existingProduct.setQuantity(productDetails.getQuantity());
40
41     return productRepository.save(existingProduct);
42 }
43
44 @Override
45 public void deleteProduct(Long id) {
46     productRepository.deleteById(id);
47 }
48 }
```

The bottom of the IDE shows the 'Boot Dashboard' and a search bar with the text 'Type tags, projects, or working set names to match (incl. * and ? wildc'. Below the search bar, there is a 'local' button. The bottom right corner shows the 'Problems', 'Javadoc', 'Declaration', and 'Console' tabs, with the message 'No consoles to display at this time.'

StudentKlhApiApplication.java

